

L4 Anti-discrimination Legislation

🕒 Created	@October 3, 2022 10:18 AM
📁 Class	Labour Economics
📋 Notes	
🌟 Type	Final
🖨️ Print	☒

[Overview](#)

[Measures and Comparisons](#)

[Theory](#)

[Perfect Labor Markets](#)

[Employers](#)

[Colleagues](#)

[Customers](#)

[Imperfect Labor Markets](#)

[Monopsony](#)

[Statistical Discrimination](#)

[Occupational Crowding](#)

[Empirical Evidence](#)

[Gender Earnings and Employment Gaps](#)

[Blinder-Oaxaca Decomposition of the Black-White Wage Gap](#)

[Correspondence Studies: Discrimination in the Labor Market](#)

[Race](#)

[Immigrants](#)

[Sexual Orientation](#)

[Appearance](#)

[Age](#)

[Criminal History](#)

[Other Studies](#)

[Policy Issues](#)

[Debate on Equal Pay](#)

[Anonymized Applications](#)

[Against](#)

[For](#)

[Criticism](#)

[Minority Quotas](#)

[Gender Quotas in Corporate Boards](#)

[Interactions with other Institutions](#)

[Future Prospects](#)

Overview

If wages differences are related to differences in productivity, there is no economic inefficiency. However, sometimes wage differences are not related to differences in productivity → **discrimination**.

discrimination

the valuation in the marketplace of personal characteristics of the worker that are **unrelated to worker productivity**

NB It can emerge from differential market power, prejudice, lack of information, employment barriers. Its manifestations may be through employers' search for new employees, labor force participation, and earnings.

premarket vs. market discrimination

This class deals with discrimination in the labor market. Please note that discrimination occurs also before individuals enter the labor market → *e.g. educational system*. These are intertwined → an individual that expects to be discriminated upon will pursue less education, since returns to education are lower anyway.

Measures and Comparisons

All countries in the OECD have a legal and institutional framework to fight labor market discrimination on gender or ethnic grounds.

Whether these frameworks are effective depends on

1. how difficult it is for workers to take action
2. how hard it is to prove discrimination
3. how severely violations by employers are punished

workers incentives

We distinguish between

1. burden of proof to be provided by the plaintiff

presumption, strong presumption, proof

2. protection against victimization of the plaintiff

this means plaintiff protection. Limited protection refers to the case where the plaintiff is protected against wrongful dismissal only.

employer incentives

There are different types of sanctions.

1. Publicity

nominate publication of the discrimination case

2. Fines

3. Prison sentences

	Workers incentives		Employer incentives to comply		
	Burden of Proof	Protection	Publicity	Fines	Prison
Australia	Proof	Yes	Yes	Penal	Yes
Austria	Strong presumption	Yes	No	Penal, rare, low	None
Belgium	Presumption	Yes	Yes	Gender: none Ethnicity: penal, low	Gender: none Ethnicity: yes
Canada	Proof	Limited	No	None	None
Czech Republic	Strong presumption	Limited	No	Administrative	None
Denmark	Gender: presumption Ethnicity: strong presumption	Limited	No	Penal	None
Finland	Presumption	Gender: yes Ethnicity: limited	No	Penal	Yes
France	Presumption	Limited	Yes	Penal	Yes
Germany	Presumption	Yes	Yes	Administrative and penal, low	None
Greece	Presumption	Yes	Gender: yes Ethnicity: no	Administrative	Yes
Italy	Gender: strong presumption Ethnicity: proof	Gender: no Ethnicity: limited	Yes	No	None
Japan	Proof	Yes	Yes	Penal	Yes
Korea	Gender: presumption Ethnicity: proof	Gender: yes Ethnicity: limited	Yes	Penal	Yes
Mexico	Strong presumption	Limited	Yes	Labor law	Yes
Netherlands	Presumption	Limited	Yes	Penal	Yes
Norway	Presumption	Yes	No	Administrative	Gender: no Ethnicity: yes
Poland	Presumption	Limited	No	None	Yes
Portugal	Presumption	Yes	Yes	Some	None
Spain	Strong presumption	Yes	Yes	Some	Yes
Sweden	Presumption	Yes	No	None	None
Switzerland	Presumption	Limited	Yes	Some	None
United Kingdom	Strong presumption	Yes	Yes	None	None
United States	Proof	Yes	Yes	Some	None

Theory

We focus on gender-based discrimination through **wage gaps**.

Perfect Labor Markets

	Employer	Co-worker	Customer
--	----------	-----------	----------

	Employer	Co-worker	Customer
Short-run	There is discrimination, firms hire only male workers or female workers depending on their degree of discrimination. Non-prejudiced firms make higher profits.	Firms will find convenient to hire only men or only women to avoid paying a surplus to prejudiced men.	Firms will pay workers their marginal product, so they will pay women less. Segregation will occur (men firms have high prices, high wages but women firm have low prices, low wages).
Long-run	Prejudiced firms will be driven out of the market since they are inefficient.	No change	No change
Level of discrimination	Market Level <i>there is gender wage gap</i>	Firm Level <i>there is no gender wage gap</i>	Firm Level <i>there is gender wage gap</i>
Segregation	Yes, but will disappear in the long-run	Yes, may persist	Yes, may persist
Within-firm discrimination	Yes	No	No

We follow the theory of **taste-based discrimination** (i.e. discrimination based on prejudice), theorised by *Becker, 1971*.

Taste-based discrimination may be perpetuated by:

1. employers
2. co-workers
3. customers

assumptions

- labor is homogeneous
- labor markets are competitive
- all workers are equally productive
- firms and workers are wage takers → wages are determined at the market levels and cannot be influenced by anyone
- discrimination leads to $w_f < w_m$

Employers

Employers translate their dislike for women in monetary costs.

$$U^{\text{employer}} = \Pi - \omega \cdot \underbrace{w_f L_f}_{\text{wage bill of women}}$$

- L_f is the number of hired female workers
 - ω is the employer-specific coefficient of discrimination, with $0 \leq \omega \leq \omega^{\max}$
- NB** workers that are non-prejudiced have $\omega = 0$, while very prejudiced employers have $\omega = \omega^{\max}$.

labor costs

- **cost of male labor force** → $w_m L_M$
- (perceived) **cost of female labor force** → $(1 + \omega)w_f L_f$

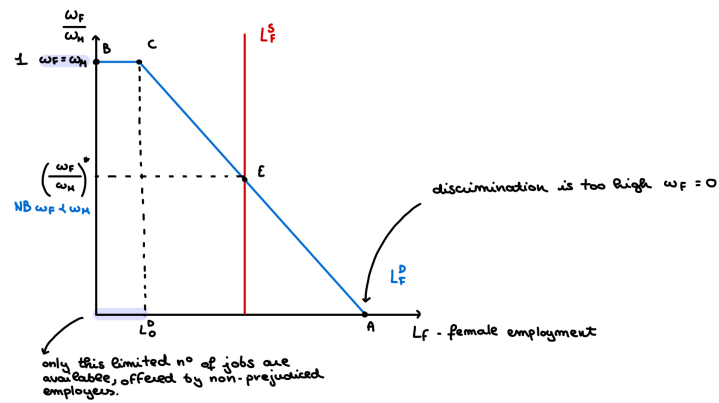
Utility-maximizing employers will hire only men if $w_m < (1 + \omega)w_f L_f$, only women otherwise. The labor force is thus **segregated**.

Holding w_f fixed, the higher ω , the lower the amount of females being hired. Because of the absence of prejudice, in equilibrium the wages of women can be lower than wages of men.

In equilibrium, the outcome of the market process is

$$\text{coefficient of market discrimination } \Omega = \frac{w_m - w_f}{w_f}$$

relative wages vs. employment



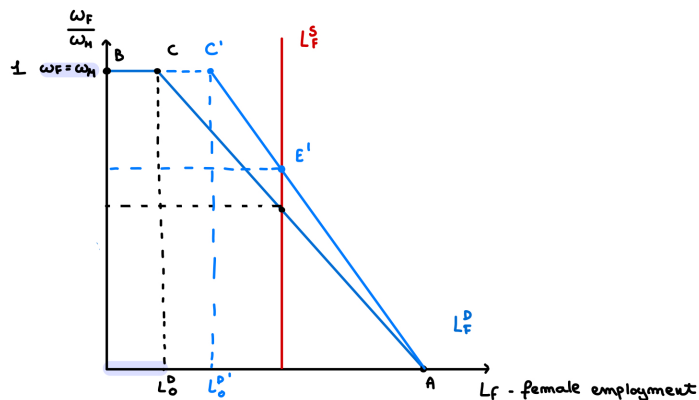
If males and females have the same wage, only the L_0^d jobs offered by unprejudiced firms are available. Female employment rises only if the relative wage goes down.

In equilibrium, all firms that employ women (**even the unprejudiced ones**) pay the same, lower, market wage $w_f^* < w_m^*$.

By analyzing shocks, we can reach another two important conclusions.

1. a larger number of employers has $\omega = 0$

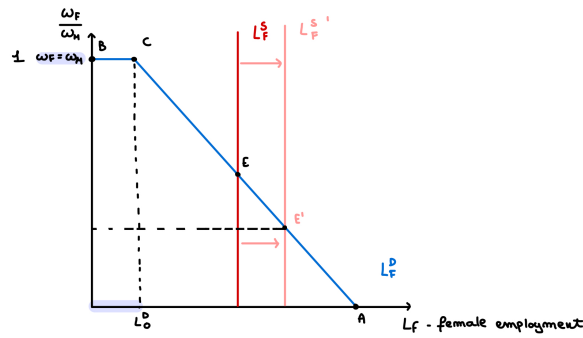
We can show that the **extent of wage discrimination depends from the marginal employer and not the average employer**. If we have an additional non-prejudiced employer, the gender wage gap will be lower.



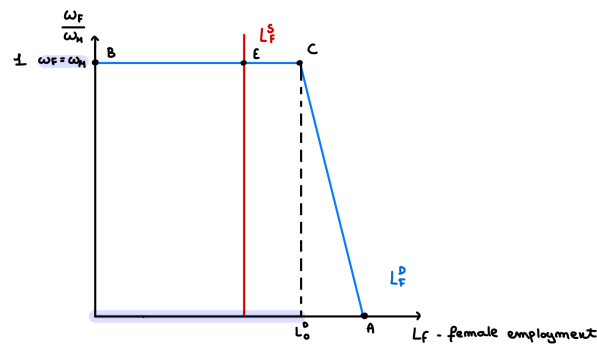
The inframarginal employer has $(\frac{w^f}{w^m})^* = \omega$ [see following section]

2. an increase in the labor supply L_F^S

The gender wage gap would increase because to provide a job to these women you need to involve employers that dislike them more.



3. there is not always a wage effect of discrimination



There is no wage effect of discrimination if $L_0^D > L_f^S$.

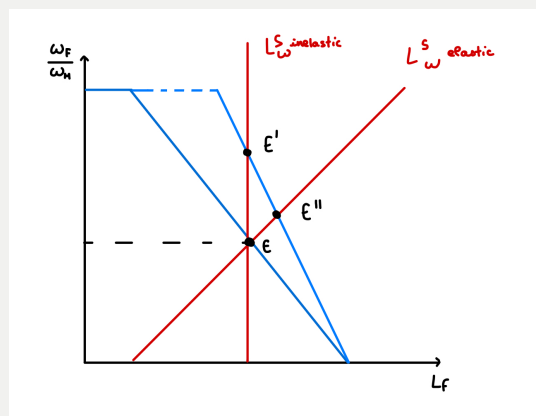
NB Here, the female labor supply is assumed to be inelastic.



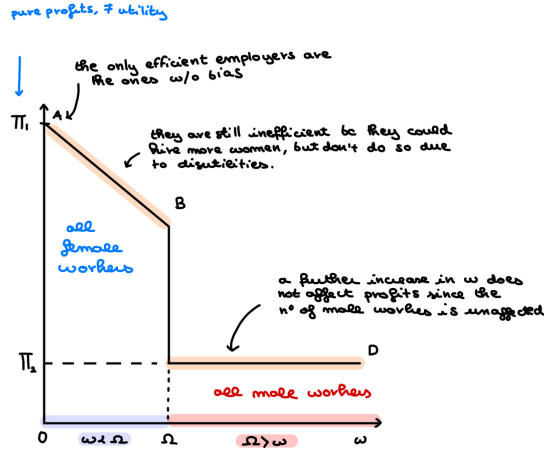
Elastic Labor Supply

The results might appear counter-intuitive - but one might guess why that is.

In this case women choose when to enter $\rightarrow$ when the shock occurs, wages increase, more women enter, which leads to a downward pressure on the wage. That is why shocks that reduce the gap reduce it less than in the inelastic case.



profits vs. discrimination coefficients



- There is a relationship between profits and coefficient of discrimination.
 - $\omega < \Omega$ or $w_m > (1 + \omega)w_f \rightarrow$ the firm will hire only female workers.
 - $\omega > \Omega$ or $w_m < (1 + \omega)w_f \rightarrow$ the firm will hire only male workers

Hiring female workers, which are cheaper, brings higher profits: the higher ω , the lower Π .

long-run

In the long-run, $\pi \rightarrow 0$ due to competition. To optimize profits, it is sensible to hire only women, since they cost less at the equilibrium. **Competition drives out discrimination, since it drives out employers that are inefficient because of their biases.**

Colleagues

Discrimination against women can arise from male coworkers. The utility that a prejudiced male gets from his wage is

$$U_m = \omega_m(1 - \delta I_f)$$

- I_f is a dummy variable for the presence of women $\rightarrow 1$ if the firm has hired at least one woman.
- ω is the coefficient of coworker discrimination.

assumptions

- All men are prejudiced against women
- Men will want to be compensated for working with women. Alternatively, for a given wage, they will prefer to work in a male firm.

conclusions

1. Firms will be segregated. In presence of women and prejudiced men working, all male workers must be paid a surplus offsetting their negative prejudices $\rightarrow$ firms hire either men or women.
2. Since firms are segregated and if employers are not prejudiced, there will be no gender wage gap even if all men are prejudiced (in the short run too).

Discrimination is present in the long-run equilibrium since there is no competitive market mechanism to drive out these firms.

Customers

Customers' willingness to pay p_w for a product may depend on the presence of female workers in its production process. We have

$$p_w = p(1 + \omega I_f)$$

If prices are equal, prejudiced customers will only buy from firms that have no female workers. They will buy from firms that have female workers if the price is low enough.

conclusions

1. Firms pay workers according to their marginal product → **in an environment with customer discrimination women will have a lower wage.**
2. Firms will be segregated. For an all women firm, the product price is low → **the firm cannot afford to hire a male worker.** For an all men firm, the product price is high, but the product price would fall if a woman is hired.
3. There is no within-firm discrimination since firms are segregated, but there is a wage gap.

Discrimination is present in the long-run equilibrium since there is no competitive market mechanism to drive out these firms.

Imperfect Labor Markets

Taste-based discrimination may occur also in imperfect labor markets (and most importantly, employer taste-based discrimination can persist in the long-run).

Other types of discrimination in a non-competitive labor market include:

1. Monopsony
2. Statistical discrimination
3. Occupational Crowding

Monopsony

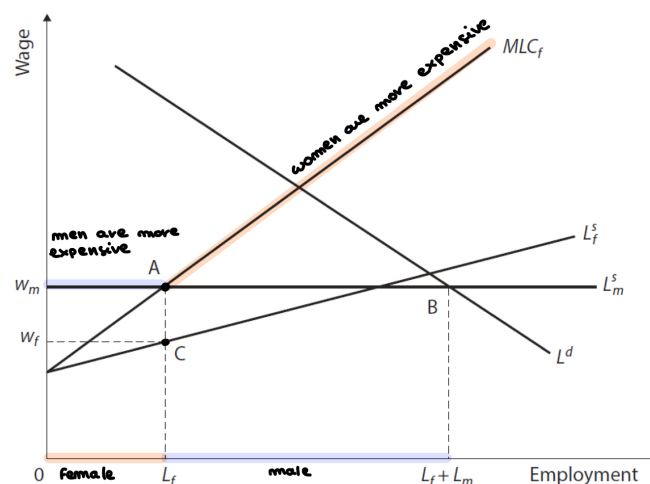
We will analyze *Robinson, 1933* theory of monopsony to explain the gender pay gap. **Employers may have more monopsony power over women than they do have over men.**

assumptions

- men and women are equally productive
- the labor market for men is fully competitive
- the labor market for women is monopsonistic

Women have higher mobility costs, thus on aggregate they are less responsive to change in wages because it's harder for them to drop everything and look for another job. Their labour supply is thus upward sloping.

model



The total employment is determined by the intersection of L^D and of the perfectly elastic labor supply of men L_M^S , which is the perfectly competitive and efficient level of employment.

- Since the supply of male labor is perfectly elastic, it means that the marginal cost is fixed and always the same, w_m . This is not the case for the marginal cost of female labor, which is described by MLC_f .
- Employers will hire women as long as $MLC_f \leq w_m$. This occurs at point *A*: the corresponding level of employment is L_f , and thus the corresponding wage is w_f , which is lower than w_m .

The gender wage gap thus originates from women having a more inelastic labour supply.

critics

This theory was not considered relevant because empirically women have a more elastic labour supply than men. However, this theory is still valid. There are two distinct matters:

1. one thing is to talk about the individual decision of a woman that has decided to work

in this case, employers have monopsony power → Robinson, 1933. Hirsch et al., 2010 also find that women's labor supply to German firms is less elastic than men's.

2. another is to talk about decision to enter the labor market

here, it might be that at the market level, labor supply is more elastic than men's. Manning, 2003 proved that, for women, market level labor supply is more elastic than at the individual firm level.

To deal with this type of discrimination → kill mobility costs.

Statistical Discrimination

Individuals are identical from the point of view of the sought characteristic but they belong to different groups.

assumptions

- **informational asymmetries**

employers have imperfect information about the true productivity of individual workers. They get information through individual test scores (e.g. standardized tests, CVs, application letters, past experience). Because test scores are imperfect, test scores are combined about information on the group the individual applicant belongs to.

- **form of perceived productivity**

The perceived productivity q of individual i in group j amounts to the weighted average of the individual test score and the group test score (i.e. the perceived group productivity).

$$q_{ij} = \alpha_j T_j + (1 - \alpha_j) T_i$$

- q_{ij} → perceived performance of the individual to the employer
- α → measure of imprecision of the signal provided by the test score, $0 < \alpha < 1$

NB when $\alpha = 0$ the signal is perfect → group performance is not informative.

- T_i → performance of the individual
- T_j → average performance of the group [it is also the average of the q_{ij}]

On average, the test score is equal to perceived productivity.

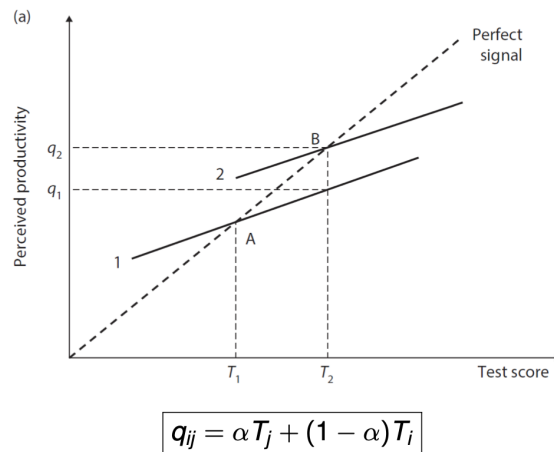
$$\sum_i q_{ij} = q_j = \alpha T_j + (1 - \alpha) \sum_i T_i = T_j$$

model

There are two types of statistical discrimination.

1. **Different perception of group productivity**

Test scores have the same accuracy for both groups, but there are different perception of group productivities → $\alpha_i = \alpha_k$ but $T_j \neq T_k$. Individuals having the exact same characteristics but belonging to different groups may get treated differently.



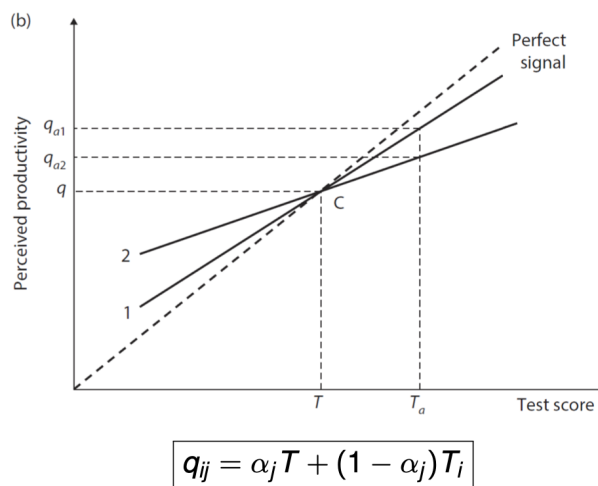
- The two lines are parallel → the test score has the same accuracy for both groups.
- The mean productivities are different → at the same test score, there is a difference in perceived productivity.

NB To the extent that the perceived group differences are real, there will be no discrimination as, on average, perceived productivity coincides with actual productivity. However, it is possible that perceived differences are based on prejudice or lack of information.

- **stereotyping** → certain perceived characteristics of a minority group are attributed to individuals from that group.

2. Different trust in provided signals

Individual test score of one group is valued more than one of another group, even though average productivity is perceived to be the same → $\alpha_i \neq \alpha_k$ but $T_i = T_k = T$.



- If you are in the group where the signal is weaker, you are better off by performing worse than the average of the two groups.
- If you are in the group where the signal is stronger, you are better off by performing better than the average of the two groups.

NB on average there is no discrimination.

Occupational Crowding

The wage discrimination analyzed by occupational crowding is not within occupations but between occupations (male-dominated jobs vs. female-dominated jobs).

occupational crowding

discrimination that arises when some groups of workers are restricted in their entrance to certain occupations. Barriers to entry could come from unions, customs, or self-selection.

assumptions

1. after being banned from male-dominated jobs, women still want to work and will go to female dominated jobs
2. men working in female jobs do not respond to the wage differential by moving to male jobs

model

If women are not allowed to enter occupation X , they will enter occupation Y , pushing the wages for occupation Y down. *e.g. marriage bars, which prohibited employment of married women.*

1. No occupational barriers

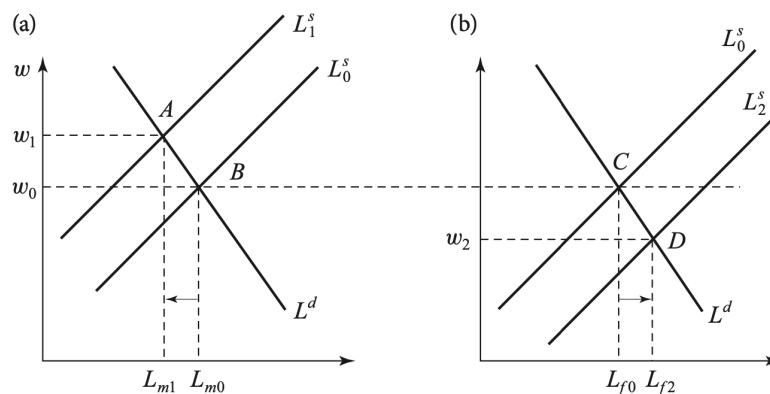
The labor markets are in equilibrium at wage w_0 . The labels “male-dominated” and “female-dominated” do not have any labor market implication → both genders can find employment in both fields.

2. Barriers are introduced

If women are forbidden to enter “male-dominated” jobs:

- i. the supply for “male-dominated” jobs will go down and wages will increase to w_1
- ii. the supply for “female-dominated” jobs will increase and wages will go down to w_2

NB wage differentials are based on the type of job, not on the gender of the worker. On average, women will earn less than men, but within industries there is no differential between genders.



Discrimination is inefficient from an economic point of view (and this holds everywhere). In the occupational crowding example.

- the market surplus in male-dominated jobs goes down (area $ABL_{m1}L_{m0}$) while the market surplus in the female-dominated jobs goes up (area $CDL_{f2}L_{f0}$). The decrease in the surplus in male-dominated jobs is greater than the increase in the female-dominated ones → net surplus decreases.
- because of the higher male wages, too little is produced in male jobs and too much in female jobs.

Empirical Evidence

There are two types of approaches.

1. traditional approach

The traditional approach looks at residual wage differentials, which are obtained by estimating multivariate wage equations. The portion of the wage differential that can't be attributed to observable characteristics is regarded as discrimination.

This has problems → it may understate as well as overstate the extent of wage discrimination (Altonji, Blank 1999).

- a. Some of the observed variables may be affected by discrimination → *e.g. women invest less in education because they anticipate lower returns.*

- b. The regression analysis may suffer from omitted variables that are related to human capital variables and personal tastes → *e.g. bias*

2. quasi-experimental approach

Two types of studies have been carried out.

a. audit studies

Fictitious candidates interview for jobs; they differ only for characteristics of interests.

Problems with this approach include:

- i. It can be very expensive
- ii. It can be hard to obtain adequately diversified fictitious candidates
- iii. Hawthorne effect → fictitious candidates could exhibit different attitudes since they know they are not really competing for the job.

b. correspondence studies

Introduction of fictitious CVs and cover letters, with the aim of studying callback rates.

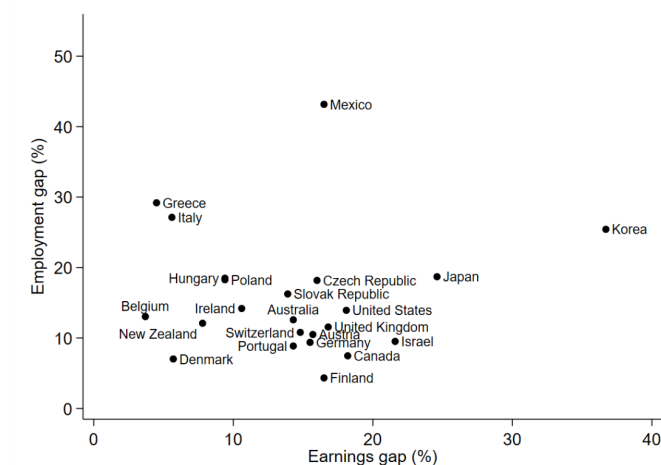
Problems with this approach include:

- i. ethical issues → employers are wasting time on fake applications.
- ii. callback rates may be influenced by unobservable characteristics
- iii. if real candidates know which firms discriminate more and avoid them, discrimination can be overestimated.
- iv. they may reveal discrimination at the invitation stage, but this is a poor predictor of the actual hiring → *Cahuc et al., 2019*

NB Piketty has proposed to build an observatory in France to systematically analyze discrimination using this method.

Gender Earnings and Employment Gaps

Country	Employment Gap 2017 (%)	Median Earnings Gap 2016 (%)	Change in gap (%)		Share of women boards large firms 2019	Gender quota 2019
			Employment, 2000-2017	Median wage, 2000-2015		
Australia	13	14	-38	-24	31	No
Austria	10	16	-55	-26	31	Yes
Belgium	13	4	-50	-65	36	Yes
Canada	7	18	-46	-22	29	No
Chile	27	—	-45	—	8	No
Czech Republic	18	16	-20	18	9	No
Denmark	7	6	-38	—	30	Yes
Estonia	9	—	-20	—	9	No
Finland	4	17	-49	-11	34	No
France	10	—	-48	—	45	Yes
Germany	9	16	-54	-19	36	Yes
Greece	29	5	-30	—	10	No
Hungary	18	9	-12	-33	13	No
Iceland	5	—	-33	—	46	Yes
Ireland	14	11	-52	-27	26	No
Israel	10	22	-51	—	22	Yes
Italy	27	6	-35	—	36	Yes
Japan	19	25	-37	-24	8	No
Korea	25	37	-19	-11	3	No
Latvia	5	—	-59	—	32	No
Lithuania	1	—	-87	—	12	No
Luxembourg	11	—	-68	—	13	No
Mexico	43	17	-17	—	8	No
Netherlands	11	—	-50	—	34	Yes
New Zealand	12	8	-36	10	38	No
Norway	4	—	-54	-30	40	Yes
Poland	18	9	-9	—	24	No
Portugal	9	14	-57	—	25	No
Slovak Republic	16	14	-6	—	29	No
Slovenia	9	—	-29	—	25	No
Spain	16	—	-61	—	26	Yes
Sweden	4	—	-31	—	38	No
Switzerland	11	15	-48	—	25	No
Turkey	54	—	-14	—	18	No
United Kingdom	12	17	-31	-35	33	No
United States	14	18	-12	-18	26	No



trends

- Generally, there has been a decline in the employment gap from 2000 to 2017.
- Information on the evolution of the earnings gap is limited. Except from New Zealand, there were substantial reductions in the median earnings gap.
- If we analyze the association between the employment gap and the gender wage gap, we find a positive association. The bulk of countries is concentrated in similar gaps of 10 to 20%.
- Some countries that have a large employment gap have low earnings gaps (Greece, Italy)
 - Women have a lower employment rate; you tend to have more educated women in the employed women spectrum, since lower-educated women do not work. Selection is stronger in women → **cream-skimming**.
 - Unconditional wage gaps thus lead to misleading results → we must take into account individual worker characteristics → **Blinder-Oaxaca Decomposition of the Wage Gap**

NB If you account for individual worker characteristics, Finland has a smaller earnings gap than Italy or Greece.

Blinder-Oaxaca Decomposition of the Black-White Wage Gap

To establish the extent of wage discrimination, one can estimate separate wage equations → it **allow us to see if there are discriminations**. Indeed, we take into account personal and jobs characteristics, but also what are the differences in how these characteristics are rewarded.

NB The higher the number of characteristics analyzed, the less precise the results.

$$\log \bar{w}_w - \log \bar{w}_b = \bar{x}_w \beta_w - \bar{x}_b \beta_b$$

where $\bar{w}$ are average wages, $\bar{x}$ vectors of average personal characteristics, and β vectors of coefficients.

According to the decomposition, the wage gap between two groups of workers can be split in three components.

$$\log \bar{w}_w - \log \bar{w}_b = \underbrace{(\bar{x}_w - \bar{x}_b) \beta_b}_{\text{diff. in characteristics}} + \underbrace{\bar{x}_b (\beta_w - \beta_b)}_{\text{diff. in coefficients}} + \underbrace{(\bar{x}_w - \bar{x}_b) (\beta_w - \beta_b)}_{\text{interaction term}}$$

- *diff. in characteristics* → including all relevant characteristics influences whether the discrimination part of the wage gap is measured correctly. $(\bar{x}_w - \bar{x}_b)$ may also be due to discrimination → if I know I will be discriminated against, I will study less.
- *diff. in coefficients* → different reward for the same characteristics. **It is the direct measure of discrimination.**

empirics

Carruthers and Wanamaker, 2017) took wage data from 10 southern US states, taking into account differences in school quality, years of schooling, and local characteristics.

Their decomposition is the following. Differences in characteristics are dominant, and the study relates them to **differences in school quality** → school resources varied across races in first half

20th century.

Characteristics	-0.311
Coefficients	-0.104
Interaction	-0.114
Total	-0.529

With equal school quality, they estimate wage inequality would have been reduced by 30-50%.

Correspondence Studies: Discrimination in the Labor Market

Race

- **There is race discrimination in hiring.**

Bertrand and Mullainathan, 2004 conducted a correspondence study by sending similar resumes, but some with white-sounding names and other with african-american sounding names, to the same job posting. Resumes with white names had a 10% callback rate, while the ones with black names had 6% chance.

Immigrants

- **There is discrimination between natives and immigrants.**

Carlsson and Rooth, 2007 investigated the presence of discrimination between natives and immigrants in hiring in Sweden. Applications with the same skills were randomly assigned Swedish (29% callback rate) or Middle-eastern names (20% callback rate).

They found that female recruiters are less likely to discriminate, while small firms and/or firms that are located in municipalities with many immigrants are more likely to discriminate.

- **Discrimination may vanish in hard-to-fill vacancies.**

Baert et. al, 2016 investigate whether discrimination between natives and immigrants is affected by the state of the labour market. CVs are assigned Flemish or Turkish sounding names. In hard-to-fill vacancies, callback rates were similar. In easy-to-fill vacancies, Flemish callback rates were about twice as high as for Turkish.

Sexual Orientation

- **There is discrimination against queer people**

Ahmed et al., 2013 investigated whether homosexuals experienced discrimination in the hiring process in Sweden. They indirectly mentioned the sexuality of the applicant (mentioning an organization).

On average, gay men received 4% lower callback rates than heterosexual men, while for women it was 6%. Discrimination occurred only in the private sector and depended on the industry: the gay (lesbian) applicant was discriminated against only in typically male (female)-dominated fields.

Appearance

- **There is discrimination against obese people**

Rooth, 2009 digitally altered photos → similar looks but different degrees of obesity. Obese males had a 6% lower callback than normal-weighted males, while for women it was 8%.

- **There is discrimination against unattractive people**

Ruffle and Shtudiner, 2015 proved that attractive males are more likely to receive a callback than plain males, while for females the difference between the two is small.

Age

- *Neumark et al., 2019* → older women are discriminated against, while there is less evidence for men.
- *Baert et al., 2016* → it is difficult to study age discrimination because subjects have different levels of experience. They conducted a correspondence study randomizing i) the age of the applicants ii) what the older applicants had done in their additional years of experience (in-field employment, out-of-field employment, inactivity). Older workers with in-field employment did not experience age discrimination.

Criminal History

Agan and Starr, 2018 studied the effect of the introduction of **ban-the-box policies** → employers do not ask about an applicant's criminal history.

They set up a correspondence study sending applications to entry-level jobs in NJ and NY, randomizing black and white names.

After BTB, black applicants were 3.8% less likely to receive a callback rate →

employers were making race-based assumptions about who is likely to have had a felony conviction.

	Callback rates (%)		
	Before BTB		After BTB
	Crime	No crime	
White applicant	8.7	14.1	14.8
Black applicant	8.0	13.4	10.3
Δ	0.7	0.7	4.5
ΔΔ			3.8

Other Studies

manager bias

Glover et al., 2017 studied the performance of cashiers in relation to their managers. When minority cashiers work with managers that are biased (i.e. associate the minority with poorer performance), their productivity is lower. This is because they interact less with their managers and therefore exert less effort.

Lower productivity of minority cashiers generates statistical discrimination in hiring.

Policy Issues

Debate on Equal Pay

equal pay

Equal pay legislation implies that men and women doing the same job should be paid the same.

- Some people argue that this protects discriminating employers. Since they're forced to pay women the same as men, the long-run mechanism in which they're driven out of business does not work anymore → they won't be penalized much, also because equal work is hard to measure.

pay transparency

Pay transparency has been introduced in some countries to fight pay gaps.

- Bennendsen et al., 2020 → in Denmark, in 2016, pay transparency for all firms with more than 35 employees decreased the gap by 7%.

Anonymized Applications

These are useless in perfect labor markets, because of the competition mechanism. We prove that they are not useful in case of imperfect labor markets. Indeed, it is not clear whether they are beneficial for minority groups.

Against

anonymized resumes for minority groups study

Behagel, Crepon and Le Barbanchon, 2015 study the effects of a 2010 field experiment on prescreened applicants about whom the French public employment agency would inform firms if they had a vacancy. Firms that were willing to participate would randomly receive standard service or the anonymous resumes.

Participating firms interviewed and hired more minority candidates with standard resumes than they did with anonymized resumes.

	Interview Rate		Hiring Rate	
	Minority	Majority	Minority	Majority
Standard	9.3	11.6	2.3	2.1
Anonymous	4.7	17.7	1.7	5.2
Δ	-4.6	6.1	-0.6	3.1
ΔΔ	-10.7		-3.7	

This is because anonymized resumes prevented them to treat minority candidates more favourably → if you're a minority, resume gaps or other flaws are interpreted in light of the circumstances.

anonymized resumes for economics PhD study

Krause et al., 2012 found that with anonymized resumes, women applying for PhDs in an European-based research institutions were penalized. Indeed, the institution wanted to attract more female researchers → this advantage disappears with anonymized resumes.

For

blind auditions study

Goldin and Rouse, 2000 study the effect of the introduction of blind rounds in US orchestras → candidates could audition concealing their identity. They compare 1950-1995 data to see whether auditioning in a blind round increases the chance to be hired.

Women experienced 65% increase in the probability of being hired (there was no difference for men). One-third of the increase in the proportion of females among new hires can be attributed to the introduction of the blind hiring (the other two thirds to more women auditioning).

Criticism

Focusing on the first stage of the hiring process gives us little insight on potential discrimination during the overall process.

Neumark et al., 2020 → even if no age is mentioned on the CV, when later on age is revealed, older applicants face discrimination and lower hiring chances.

Minority Quotas

These are useless in perfect labor markets, because of the competition mechanism. We prove that they are useful in case of imperfect labor markets.

affirmative action policy

Fryer and Loury, 2005

It regulates the allocation of scarce positions so as to increase the representation of persons belonging to certain population subgroups. It requires proactive steps. One common criticism is that it is stigmatizing.

Gender Quotas in Corporate Boards

They are present in few countries in Europe (Italy, Portugal, Austria, Norway), with the aim to correct gender imbalances.

Pros

1. *Smith, 2018* → increased gender diversity is thought to increase the quality of corporate decisions and influence positively the gender balance in other decision making positions.
2. *Profeta, 2017* → quotas can increase competition by encouraging more women to aim for board positions, which in turn can increase the quality of board members of either gender. [but *Bennouri et al., 2020* disagrees]

Cons

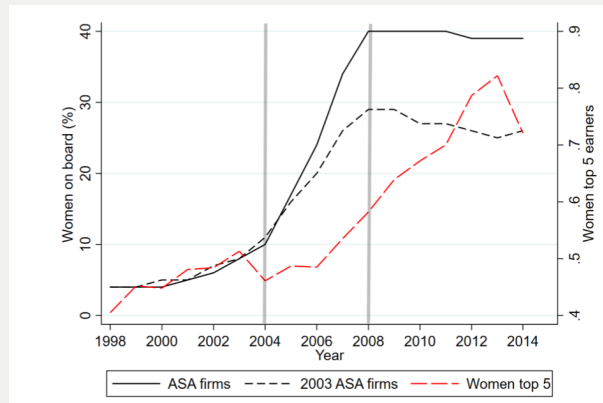
1. Gender quotas are difficult to implement and take a long time to reach.
2. Some argue that quotas impose reverse discrimination and indicate lack of qualified female candidates → the firm's overall human capital could decrease, hindering the firm's performance. However, the reverse may occur → human capital increases because positions are given to people who would otherwise be passed over.



Spillover Effects of Quotas

Norway was the first country to adopt mandatory quotas in 2006. Publicly listed companies were required to increase their share of female directors on the board to at least 40% in two years.

- *Bertrand et al., 2019* → the main results were:
 - Increase of women in the top 5 highest-earning individuals within the firms.
 - 30% increase of women in boards (some ASA firms unlisted to avoid meeting the quota).
 - The gender pay gap within the board fell substantially, however the new female board members did not affect the opportunities for other women in their firm.



- *Ahern and Dittmar, 2012* → new female directors are younger and more educated [similar result in Italy, with *Ferrari et al., 2018*]
- *Matsa and Miller, 2013* → possible changes in corporate strategies (fewer layoffs)

Interactions with other Institutions

It interacts with:

1. education and training

- Very important in cases of pre-market discrimination, that occurs most often on ethnic grounds.
- Individuals may underinvest in education in expectation of discrimination in the future.

2. unions

- Unions help minorities and women to recover their full returns to their investment in education.
- There is not much tradition of unions fighting against discriminations.

3. family policies

- Anti-discrimination policies might reinforce family policies that aim to increase female labor force participation.

4. working hours legislation

- Legislation that discourages part-time working may affect the labor force participation of women.

5. employment protection legislation

- a worker can be reinstated if a layoff appears to be discriminatory.

Future Prospects

There are two main reasons for ADL to exist.

1. Equity

Equal pay is a human right → *Art. 23 Universal Declaration*

2. Efficiency

ADL improves labor market efficiency through better allocation of workers to jobs. The importance of ADL depends on whether the labor market is competitive.

	Competitive	Imperfect
Dynamics	Discriminatory employers will be punished irrespective of whether the discrimination comes from prejudiced employers or from statistical discrimination using stereotyping.	Some of the market power may be used to sustain discrimination against women or minority groups.
Employers	The marginal employer determines the degree of discrimination.	The average employer determines the degree of discrimination.

direct effects vs. feedback effects of discrimination

Direct discrimination is related to differences in wages, positions, that are unrelated to productivity differences. Indirect discrimination are reactions to direct discrimination → if I am discriminated, I may invest less in education.

Blau et al., 2010 suggest that only 40% of the wage difference between men and women is explained by direct effects. Direct effects of discrimination will be a lower bound of the total effect.
